Advance Python Assignment

Topic-10: Search and Match Functions

Q1. Explain re.search() function in Python.

Ans.
The re.search() function in Python is used to find a pattern anywhere in a string. It checks the entire string and returns the first match if the pattern exists. If no match is found, it returns None. This function is part of the re module, which is used for working with regular expressions. It is helpful when we want to check if a specific word or pattern is present in a string.

Example:
```
import re

sentence = "Learning Python is fun"
result = re.search("Python", sentence)

if result:
   print("Pattern found")
else:
   print("Pattern not found")
```

Q2. Explain re.match() function in Python.

Ans.
The re.match() function in Python is used to match a pattern only at the beginning of a string. It returns a match object if the pattern is found at the start, otherwise it returns None. Unlike re.search(), it does not check the entire string. This function is useful when we want to validate patterns that must appear at the start of a string.

Example:
```
import re

text = "Python programming"
result = re.match("Python", text)
```

```
if result:
    print("Match at beginning")
else:
    print("No match")
```